Determinant & Inverse

Tuesday, 3 February 2026 12:15 pm

Matrix Algebra

Definition 9.1: Given matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix},$$

both of the same dimension $m \times n$, the sum $\mathbf{A} + \mathbf{B}$ is defined as

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}.$$

Definition 9.2: For a scalar α we define $\alpha\mathbf{A}$ by

$$\alpha\mathbf{A} = \alpha \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{bmatrix}.$$

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Example 9.3. Given $\mathbf{A}$ and $\mathbf{B}$ below, find $3\mathbf{A} - 2\mathbf{B}$.

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 5 \\ 0 & -3 & 9 \\ 4 & -6 & 7 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 5 & 0 & -11 \\ 3 & -5 & 1 \\ -1 & -9 & 0 \end{bmatrix}$$

Matrix Multiplication

Definition 9.5: For $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times p}$, with $\mathbf{B} = \begin{bmatrix}\mathbf{b}_1 & \mathbf{b}_2 \cdots & \mathbf{b}_p\end{bmatrix}$, we define the product $\mathbf{AB}$ by the formula

$$\mathbf{AB} = \begin{bmatrix}\mathbf{Ab}_1 & \mathbf{Ab}_2 & \cdots & \mathbf{Ab}_p\end{bmatrix}.$$

Example 9.6. For $\mathbf{A}$ and $\mathbf{B}$ below compute $\mathbf{AB}$ and $\mathbf{BA}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -4 & 2 & 4 & -4 \\ -1 & -5 & -3 & 3 \\ -4 & -4 & -3 & -1 \end{bmatrix}$$

Example 9.7. For $\mathbf{A}$ and $\mathbf{B}$ below compute $\mathbf{AB}$ and $\mathbf{BA}$.

$$\mathbf{A} = \begin{bmatrix} -4 & 4 & 3 \\ 3 & -3 & -1 \\ -2 & -1 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1 & -1 & 0 \\ -3 & 0 & -2 \\ -2 & 1 & -2 \end{bmatrix}$$

If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is a square matrix, the kth power of $\mathbf{A}$ is

$$\mathbf{A}^k = \underbrace{\mathbf{AAA}\cdots\mathbf{A}}_{k \text{ times}}$$

Example 9.9. Compute $\mathbf{A}^3$ if

$$\mathbf{A} = \begin{bmatrix} -2 & 3 \\ 1 & 0 \end{bmatrix}$$

Matrix Transpose

Definition 9.10: Given a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, the **transpose** of $\mathbf{A}$ is the matrix $\mathbf{A}^T$ whose ith column is the ith row of $\mathbf{A}$.

If $\mathbf{A}$ is $m \times n$ then $\mathbf{A}^T$ is $n \times m$. For example, if

$$\mathbf{A} = \begin{bmatrix} 0 & -1 & 8 & -7 & -4 \\ -4 & 6 & -10 & -9 & 6 \\ 9 & 5 & -2 & -3 & 5 \\ -8 & 8 & 4 & 7 & 7 \end{bmatrix}$$

then

$$\mathbf{A}^T = \begin{bmatrix} 0 & -4 & 9 & -8 \\ -1 & 6 & 5 & 8 \\ 8 & -10 & -2 & 4 \\ -7 & -9 & -3 & 7 \\ -4 & 6 & 5 & 7 \end{bmatrix}.$$

Compute $(\mathbf{AB})^T$ and $\mathbf{B}^T\mathbf{A}^T$ if

$$\mathbf{A} = \begin{bmatrix} -2 & 1 & 0 \\ 3 & -1 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -2 & 1 & 2 \\ -1 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

Theorem 9.12: Let $\mathbf{A}$ and $\mathbf{B}$ be matrices of appropriate sizes. The following hold:

(1) $(\mathbf{A}^T)^T = \mathbf{A}$
(2) $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$
(3) $(\alpha\mathbf{A})^T = \alpha\mathbf{A}^T$
(4) $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$

Inverse of a Matrix

Definition 10.2: If $\mathbf{A}$ is invertible then we denote **the** inverse of $\mathbf{A}$ by $\mathbf{A}^{-1}$. Thus, $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$.

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Example 10.3. Given $\mathbf{A}$ and $\mathbf{C}$ below, show that $\mathbf{C}$ is the inverse of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 1 & -3 & 0 \\ -1 & 2 & -2 \\ -2 & 6 & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -14 & -3 & -6 \\ -5 & -1 & -2 \\ 2 & 0 & 1 \end{bmatrix}$$

Theorem 10.4: Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and suppose that $\mathbf{A}$ is invertible. Then for any $\mathbf{b} \in \mathbb{R}^n$ the matrix equation $\mathbf{Ax} = \mathbf{b}$ has a unique solution given by $\mathbf{A}^{-1}\mathbf{b}$.

Example 10.5. Use the result of **Example 10.3.** to solve the linear system $\mathbf{Ax} = \mathbf{b}$ if

$$\mathbf{A} = \begin{bmatrix} 1 & -3 & 0 \\ -1 & 2 & -2 \\ -2 & 6 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix}.$$

Theorem 10.6: Let $\mathbf{A}$ and $\mathbf{B}$ be invertible matrices. Then:

(1) The matrix $\mathbf{A}^{-1}$ is invertible and its inverse is $\mathbf{A}$:

$$(\mathbf{A}^{-1})^{-1} = \mathbf{A}.$$

(2) The matrix $\mathbf{AB}$ is invertible and its inverse is $\mathbf{B}^{-1}\mathbf{A}^{-1}$:

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}.$$

(3) The matrix $\mathbf{A}^T$ is invertible and its inverse is $(\mathbf{A}^{-1})^T$:

$$(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T.$$

Computing the Inverse of a Matrix

$$\begin{bmatrix} \mathbf{A} & \mathbf{I}_n \end{bmatrix} \sim \begin{bmatrix} \mathbf{I}_n & \mathbf{c}_1 & \mathbf{c}_2 \cdots & \mathbf{c}_n \end{bmatrix}$$

Example 10.7. Find the inverse of $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ -1 & -2 \end{bmatrix}$ if it exists.

$$\begin{bmatrix} \mathbf{A} & \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 1 & 0 \\ -1 & -2 & 0 & 1 \end{bmatrix}$$

Add rows R_1 and R_2:

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ -1 & -2 & 0 & 1 \end{bmatrix} \xrightarrow{R_1+R_2} \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Perform the operation $\xrightarrow{-3R_2+R_1}$:

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \xrightarrow{-3R_2+R_1} \begin{bmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Thus, $\texttt{rref}(\mathbf{A}) = \mathbf{I}_2$, and therefore $\mathbf{A}$ is invertible. The inverse is

$$\mathbf{A}^{-1} = \begin{bmatrix} -2 & -3 \\ 1 & 1 \end{bmatrix}$$

Verify:

$$\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} 1 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} -2 & -3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Example 10.8. Find the inverse of $\mathbf{A} = \begin{bmatrix} 1 & 0 & 3 \\ 1 & 1 & 0 \\ -2 & 0 & -7 \end{bmatrix}$ if it exists.

Solution. Form the augmented matrix $\begin{bmatrix} \mathbf{A} & \mathbf{I}_3 \end{bmatrix}$ and row reduce:

$$\begin{bmatrix} 1 & 0 & 3 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ -2 & 0 & -7 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-R_1+R_2,\ 2R_1+R_2} \begin{bmatrix} 1 & 0 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 1 \end{bmatrix}$$

$-R_3$:

$$\begin{bmatrix} 1 & 0 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 1 \end{bmatrix} \xrightarrow{-R_3} \begin{bmatrix} 1 & 0 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & -1 \end{bmatrix}$$

$3R_3 + R_2$ and $-3R_3 + R_1$:

$$\begin{bmatrix} 1 & 0 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & -1 \end{bmatrix} \xrightarrow{3R_3+R_2,\ -3R_3+R_1} \begin{bmatrix} 1 & 0 & 0 & 7 & 0 & 3 \\ 0 & 1 & 0 & -7 & 1 & -3 \\ 0 & 0 & 1 & -2 & 0 & -1 \end{bmatrix}$$

Therefore, $\texttt{rref}(\mathbf{A}) = \mathbf{I}_3$, and therefore $\mathbf{A}$ is invertible. The inverse is

$$\mathbf{A}^{-1} = \begin{bmatrix} 7 & 0 & 3 \\ -7 & 1 & -3 \\ -2 & 0 & -1 \end{bmatrix}$$

Verify:

$$\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 & 3 \\ 1 & 1 & 0 \\ -2 & 0 & -7 \end{bmatrix} \begin{bmatrix} 7 & 0 & 3 \\ -7 & 1 & -3 \\ -2 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example 10.9. Find the inverse of $\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -2 \\ -2 & 0 & -2 \end{bmatrix}$ if it exists.

Solution. Form the augmented matrix $\begin{bmatrix} \mathbf{A} & \mathbf{I}_3 \end{bmatrix}$ and row reduce:

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ -2 & 0 & -2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-R_1+R_2,\ 2R_1+R_2} \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 \end{bmatrix}$$

Example 10.11. Without doing any arithmetic, write down the inverse of the dilation matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}.$$

Example 10.12. Without doing any arithmetic, write down the inverse of the rotation matrix

$$\mathbf{A} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}.$$

Determinants

Definition 11.1: Given a 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

we define the **determinant** of $\mathbf{A}$ as

$$\det \mathbf{A} = \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$

Example 11.2. Compute the determinant of $\mathbf{A}$.

(i) $\mathbf{A} = \begin{bmatrix} 3 & -1 \\ 8 & 2 \end{bmatrix}$

(ii) $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix}$

(iii) $\mathbf{A} = \begin{bmatrix} -110 & 0 \\ 568 & 0 \end{bmatrix}$

Definition 11.3: Let $\mathbf{A}$ be a 3×3 matrix. Let $\mathbf{A}_{jk}$ be the 2×2 matrix obtained from $\mathbf{A}$ by deleting the jth row and kth column. Define the **cofactor** of a_{jk} to be the number $C_{jk} = (-1)^{j+k} \det \mathbf{A}_{jk}$. Define the **determinant** of $\mathbf{A}$ to be

$$\det \mathbf{A} = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & \mathbf{a_{22}} & \mathbf{a_{23}} \\ a_{31} & \mathbf{a_{32}} & \mathbf{a_{33}} \end{bmatrix} \longrightarrow \mathbf{A}_{11} = \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}.$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ \mathbf{a_{21}} & a_{22} & \mathbf{a_{23}} \\ \mathbf{a_{31}} & a_{32} & \mathbf{a_{33}} \end{bmatrix} \longrightarrow \mathbf{A}_{12} = \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ \mathbf{a_{21}} & \mathbf{a_{22}} & a_{23} \\ \mathbf{a_{31}} & \mathbf{a_{32}} & a_{33} \end{bmatrix} \longrightarrow \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$C_{12} = (-1)^{1+2} \det \mathbf{A}_{12} = -\det \mathbf{A}_{12}$$

$$C_{13} = (-1)^{1+3} \det \mathbf{A}_{13} = \det \mathbf{A}_{13}.$$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Example 11.4. Compute the determinant of the matrix

$$\mathbf{A} = \begin{bmatrix} 4 & -2 & 3 \\ 2 & 3 & 5 \\ 1 & 0 & 6 \end{bmatrix}$$

Corollary 11.8: If $\mathbf{A}$ has a row or column containing all zeros then $\det \mathbf{A} = 0$.

Corollary 11.9: For any square matrix $\mathbf{A}$ it holds that $\det \mathbf{A} = \det \mathbf{A}^T$.

Example 11.10. Compute the determinant of

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 0 & -2 \\ 1 & 2 & -2 & -1 \\ 0 & 0 & 2 & 1 \\ -1 & -3 & 1 & 0 \end{bmatrix}$$

Theorem 11.13: The determinant of a triangular matrix is the product of its diagonal entries.

Evaluating a Determinant using Elementary Row Operations

Row Operation	Effect on Determinant
Interchange two rows	Change the sign
Multiply a row by a constant	Multiply by the constant
Add a multiple of a row to another row	No change

Corollary 12.2: If $\mathbf{A} \in \mathbb{R}^{n\times n}$ has two rows (or two columns) that are equal then $\det(\mathbf{A}) = 0$.

Example 12.5. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by interchanging rows 2 and 4, what is $\det \mathbf{B}$?

Example 12.6. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ denote the rows of $\mathbf{A}$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by replacing row $\mathbf{a}_3$ by $3\mathbf{a}_1 + \mathbf{a}_3$, what is $\det \mathbf{B}$?

Example 12.7. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ denote the rows of $\mathbf{A}$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by replacing row $\mathbf{a}_3$ by $3\mathbf{a}_1 + 7\mathbf{a}_3$, what is $\det \mathbf{B}$?

Example 12.8. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ denote the rows of $\mathbf{A}$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by replacing row $\mathbf{a}_3$ by $4\mathbf{a}_1 + 5\mathbf{a}_2$, what is $\det \mathbf{B}$?

Theorem 12.9: A square matrix $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$.

Theorem 12.10: Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and let $\mathbf{B} = \beta\mathbf{A}$, that is, $\mathbf{B}$ is obtained by multiplying every entry of $\mathbf{A}$ by β. Then $\det \mathbf{B} = \beta^n \det \mathbf{A}$.

Example 12.11. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. What is $\det(3\mathbf{A})$?

Theorem 12.12: Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Then

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

Corollary 12.13: For any square matrix $\det(\mathbf{A}^k) = (\det \mathbf{A})^k$.

Corollary 12.14: If $\mathbf{A}$ is invertible then

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det \mathbf{A}}.$$

Example 12.15. Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be $n \times n$ matrices. Suppose that $\det \mathbf{A} = 3$, $\det \mathbf{B} = 0$, and $\det \mathbf{C} = 7$.

(i) Is $\mathbf{AC}$ invertible?
(ii) Is $\mathbf{AB}$ invertible?
(iii) Is $\mathbf{ACB}$ invertible?